Statistical Genetics

Lecture 3

Multiple Comparison Procedures

Lecture 3 Readings

- **Foulkes, A.S. (2009). Applied Statistical Genetics with R: Chapter 4**

Multiple Comparison Procedures

- **Several traditional statistical methods were described for testing for an association between a genotype and a trait.**
- **These methods generally require comparing a test statistic to its known (or simulated) distribution in order to quantify the probability of seeing what we do or something more extreme under the assumption that the null hypothesis is true.**
- **Our decision to accept or reject the null then rests on a comparison of this quantity, called the p-value, to a threshold based on a previously determined acceptable level of error.**
- **Most methods adjusting for multiple comparisons describe controlling one of two error rates: the family-wise error rate (FWER) and the false discovery rate (FDR).**

		Test	
		Non-Significant	Significant
Truth	H_0	True Negative	**Type-1 Error**
	H_A	**Type-2 Error**	True Positive

Type-1 and Type-2 errors in hypothesis testing

Multiple Comparison Procedures

- **False Discovery Rate (FDR): defined as the expected proportion of null hypotheses that are true among those that are declared significant.**

		Test		
		Non-significant	Significant	
Truth	H_0	U	V	m_0
	H_A	T	S	$m - m_0$
		$m - R$	R	m

Table 1: Errors for multiple hypothesis tests

- **Returning to the notation of the above Table and assuming R, the number of tests that are declared significant, is greater than 0, FDR is given by**

$$\boldsymbol{FDR = E\left(\frac{V}{R}\right)}$$

False Discovery Rate

- For R = 0, we define the ratio V/R as identically equal to 0 since no false rejections can occur.
- The FDR can therefore be expressed equivalently as

$$\mathrm{FDR} = \mathrm{E}\left({}^{V}/_{R}\,|\mathrm{R} > 0\right) Pr(\mathrm{R} > 0) + \mathrm{E}\left({}^{V}/_{R}\,|\mathrm{R} = 0\right) Pr(\mathrm{R} = 0)$$
$$= \mathrm{E}\left({}^{V}/_{R}\,|\mathrm{R} > 0\right) Pr(\mathrm{R} > 0)$$

- The FDR is the expected number of false (null) hypotheses among the hypotheses declared to be significant.
- There are many methods to “control” the FDR (i.e., keep it below some specified quantity.)
- The FDR is often thought of as more “powerful” than controlling the chance any false positives.
- Also, one might argue that the FDR is a more relevant quantity for some purposes.

Benjamini-Hochberg Method

- The simplest method, due to Benjamini and Hochberg, uses the p-values directly.
- Remember there are as many p-values as there are tests
- First we order the p-values from smallest to largest, and call the
- i-th smallest p-value pi, i = 1,…,m.
- Then for each i we compare pi to αi/m, where α is the value at which we want to control the FDR (α may be as high as 0.1 or 0.15)
- Then we find the biggest i for which pi is less than or equal to
- αi/m
- Once we find that biggest i, we reject that hypothesis, and all the hypotheses (tests) with smaller p-values.

Bonferroni Adjustment

- **The family-wise error rate (FWER) is defined as the probability of making at least one Type-1 error: FWER = Pr(V ≥ 1)**
- **Suppose we conduct *m* tests of hypotheses, given by $H_0^1, \ldots, H_0^m$, and each test is controlled at a level equal to α.**
- **This means that, for any single test, the probability of incorrectly rejecting the null hypothesis, the Type-1 error rate, is less than or equal to *α*.**
- **Formally, we write this as**

$$Pr\left(Reject\ H_0^i \,\middle|\, H_0^i\ true\right) < \alpha$$

for all $i = 1, \ldots, m$

- **Now let us assume that these tests are independent of one another, so that the likelihood of rejecting a test does not depend on the outcomes of other tests.**
- **If we let V be the number of true nulls that are declared significant, then the probability of incorrectly rejecting at least one null hypothesis is given by**

Bonferroni Adjustment

- The *Bonferroni* adjustment for multiple comparisons is probably the most straightforward single-step adjustment to apply.
- It involves simply using $\alpha' = \alpha/m$ in place of α for the level of each test, where m is the number of tests to be performed.
- Other single-step procedures that control FWER are Tukey's and Scheffe's methods

Family-wise Error Rate

$$
\begin{aligned}
\text{FWEC} &= Pr(V \geq 1 \mid H_0^C \text{ true}) \\
&= 1 - Pr(V = 0 \mid H_0^C \text{ true}) \\
&= 1 - \prod_{i=1}^{m} Pr(\text{do not reject } H_0^i \mid H_0^i \text{ true}) \\
&= 1 - \prod_{i=1}^{m} \left[1 - Pr(\text{reject } H_0^i \mid H_0^i \text{ true})\right] \\
&\leq 1 - \prod_{i=1}^{m} (1 - \alpha) = 1 - (1 - \alpha)^m
\end{aligned}
$$

- **Notably, if m = 1, then this equation reduces to FWEC ≤ α.**
- **However, if m = 2 independent tests are performed, each at level α = 0.05, then we are only certain that the probability of making at least one type-1 error is less than or equal to 1 - (1 – 0.05)^2 = 0:0975.**

Tukey's Adjustment

- **Tukey's studentized range test is another single-step adjustment method for testing multiple hypotheses that is useful for the comparison of means between groups.**
- **For example, suppose we fit an ANOVA model of the form**

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

- **where Y_{ij} is the measured trait for the jth individual with treatment i, μ is the overall population mean, α_i is the shift in the mean resulting from receiving treatment i, i = 1, … , m and j = 1, …, n_i.**
- **Typically, we begin by conducting an overall F-test for equality of means across the treatment groups.**
- **Formally, we test the null hypothesis given by $H_0 : \alpha_1 = \alpha_2 = \ldots = \alpha_m$.**
- **If this test is significant, indicating a departure from the null, then interest may lie in identifying the specific treatment groups that are different.**
- **That is, we may now want to test the set of "M choose 2" hypotheses, given by $H_0 : \alpha_i = \alpha_j$ for i $\neq$ j.**

Tukey's Adjustment

- Tukey's honestly significantly different (HSD) test based on the studentized range distribution is a natural approach for this setting.
- Notably, in its original formulation, this method is applicable to groups that have the same sample size, so we begin by assuming $n_i = \mathrm{n}$ for all i.
- For each null hypothesis $H_0 : \mu_i = \mu_j$, a test statistic is given by

$$t_s = \frac{\sqrt{2}\,\widehat{D}}{\sqrt{\mathrm{Var}\left[\widehat{D}\right]}} \sim q_{\mathrm{m},(\mathrm{m}\times n)-m}$$

where $\widehat{D}$ is the difference in the sample means for groups i and j. It can be shown that t_s can also be written as:

$$t_s = \frac{\sqrt{2}\,\widehat{D}}{\sqrt{MSE}} \sim q_{\mathrm{m},(\mathrm{m}\times n)-m}$$

where MSE is the mean square error from fitting the full ANOVA model.

This statistic has a studentized range distribution with m and $(\mathrm{m} \times n) - \mathrm{m}$ degrees of freedom.

Examples from Association Tests